Review: Graphing linear relationships from a table of values

1 For each of the following equations and tables of values:

i Plot the points in the table of values.

ii Is the graph of the equation linear?

a $y = x$

x	-1	0	1	2
y	-1	0	1	2

b $y = 4x$

x	-2	-1	0	1
y	-8	-4	0	4

c $y = -x$

x	0	1	2	3
y	0	-1	-2	-3

d $y = -3x$

x	-3	-2	1	0
y	9	6	3	0

e $y = \frac{x}{5}$

x	-5	0	5	10
y	-1	0	1	2

f $y = -\frac{x}{3}$

x	-3	0	3	6
y	1	0	-1	-2

2 For each of the following equations and tables of values:

i Complete the given table of values.

ii Plot the points in the table of values and sketch the graph of the equation.

a $y = -4x$

x	-1	0	1	2
y				

b $y = 3x$

x	-1	0	1	2
y				

c $y = -\frac{x}{5}$

x	-10	-5	0	5
y				

d $y = \frac{x}{2} + 5$

x	-4	-2	0	2
y				

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e $y = -2x + 1$

x	-1	0	1	2
y				

f $y = \frac{x}{2}$

x	-6	-4	-2	0
y				

g $y = -\frac{x}{4} + 5$

x	-8	-4	0	4
y				

h $y = 3x + 1$

x	-1	0	1	2
y				

3 For each of the following equations and tables of values:

i Complete the given table of values.

ii Sketch the graph of the equation.

a $y = 8x$

x	-2	-1	0	1
y				

b $y = -7x$

x	-1	0	1	2
y				

c $y = \frac{x}{5}$

x	-2	-1	0	5
y				

d $y = -\frac{x}{7}$

x	-7	-4	-3	0
y				

e $y = 9x + 1$

x	-2	-1	0	1
y				

f $y = -7x - 3$

x	-1	0	1	2
y				

g $y = \frac{x}{6} - 5$

x	-5	-3	0	6
y				

h $y = -\frac{x}{3} - 6$

x	-3	-2	-1	0
y				

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i $y = \dfrac{5x+3}{4}$

x	-3	-2	0	1
y				

j $y = -\dfrac{5x+4}{3}$

x	-2	-1	0	1
y				

4 Let the height of a candle be y centimeters. If the candle is lit, the height decreases according to the equation $y = -2t + 8$, where t is the elapsed time in minutes.

a Complete the table of values below:

Time $(t$ min$)$	0	1	2	3
Height of candle $(y$ cm$)$				

b Sketch the graph of $y = -2t + 8$.

c The height of candle and the time elapsed must be non-negative.
Using part (b), for what values of t does the equation $y = -2t + 8$ accurately model the height of the candle?

5 Let the volume of water in a tank be V liters. If the tank is initially empty, then the amount of water in the tank is given by $V = 5t$ where t is the elapsed time in minutes and 5 is the rate that the tank fills in $\mathrm{L/min}$.

a Complete the table of values below:

Time $(t$ min$)$	0	1	2	3
Volume of water $(V$ L$)$				

b Sketch the graph of $V = 5t$.

c The volume of the tank is 15 L and the time elapsed is non-negative.
Using part (b), for what values of t does the equation $V = 5t$ accurately model the volume of the tank?

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